

Remote Access, Downloads & Shell Customization

Session 12

Objectives:

- **Use SSH to connect to remote systems**
- **Download and extract content using wget, curl, tar, and unzip**
- **Customize their shell with alias, environment variables, and .bashrc modifications**

SSH: Secure Shell

What is SSH?

A cryptographic network protocol that allows secure communication between two machines, even over untrusted networks.

It is commonly used for:

- Secure remote login to servers
- Safe file transfers
- Running commands on remote systems

SSH: Secure Shell

ssh [username]@[hostname or IP address]

- [username]: your remote server username
- [hostname or IP address]: the server's hostname or IP address.

Ex: ssh [thomas@192.168.56.1](#)

SSH: Secure Shell

On Windows

- check ip from ipconfig on windows or check ip from the wifi (properties)
- Optional features -> add opensshd-server
- Try Get-Service sshd , should see running otherwise: Start-Service sshd
- *net user thomas thomaspwd /add*
- *net user thomas /active:yes*

On your VM (Linux)

- *apt install openssh-server,*
- *systemctl enable ssh*
- *systemctl start ssh*

SSH: Secure Shell

On MAC:

- Check `/etc/ssh/sshd_config` and you should see `PasswordAuthentication yes`
If not change the line and then run
`sudo launchctl stop com.openssh.sshd`
`sudo launchctl start com.openssh.sshd`
- In general, network allow sharing

SSH: Secure Shell

Password input?

SSH does not allow `--password` on the command line (for security)

Use **sshpass** only for demo/testing:

```
sshpass -p 'pwd' ssh thomas@192.168.56.1
```

Secure Copy: scp

Secure file transfer is an essential skill for Linux system administration. The **scp** command (Secure Copy Protocol) allows you to **safely copy files** over a network by using **SSH encryption**.

Unlike basic tools that send data in plain text, scp encrypts everything—your password, files, and commands—protecting them from eavesdropping or interception.

```
scp file.txt user@host:/path/  
scp -r folder/ user@host:/home/user/
```

Demo: Send a file from Linux VM to Windows scp file.txt thomas@192.168.56.1:~/Desktop/

rsync

rsync is a powerful command-line tool used to **synchronize files and directories** between two locations — locally or over a network via SSH..

It only transfers differences between source and destination — making it much faster than scp for repeated transfers or large directories.

Feature	scp	rsync
Secure transfer	✓ SSH	✓ SSH
Sync only changed files	✗ No	✓ Yes
Resume support	✗ No	✓ Yes
Directory structure	✓	✓
Compression	✗ No	✓ (-z)

rsync

Syntax:

rsync [options] source destination

Option

Meaning

-a	Archive mode: preserve permissions, timestamps, symbolic links, etc.
-v	Verbose: show progress
-z	Compress data during transfer (faster over network)
-r	Recursive: copy subdirectories
--delete	Delete files from destination that are no longer in the source

-Sync folder to remote machine:

rsync -avz my_project/ thomas@192.168.1.39:~/projects/

-Pull files from a remote machine:

rsync -avz thomas@192.168.1.39:~/projects/ .

rsync

demo:

- create two folders test1 and test2*
- create 3 files into test1*
- rsync all the files from test1 to test2*
- modify one file and redo rsync*

mkdir test1 test2

cd test1;touch file1.txt file2.txt file3.txt

rsync -av test1/ test2/

gedit file2.txt

rsync -av test1/ test2/

Downloading Files: wget and curl

wget – For simple file downloads

wget gnu.org/software/wget/manual/wget.txt

wget -O lecture.pdf <https://example.com/file123.pdf> #DL and rename

curl – For more control (headers, APIs). Working with REST APIs or inspecting HTTP headers

ex: curl -O <https://example.com/file.zip> # save file

curl <https://example.com/data.csv> -o data.csv

Downloading Files: wget and curl

```
curl -X POST -H "Content-Type: application/json" -d '{"name":"Thomas"}'  
https://api.example.com/create
```

Part

curl

-X POST

-H "Content-Type: application/json"

-d '{"name":"Thomas"}'

https://api.example.com/create

Meaning

The tool that sends the request

“I want to send data” (like submitting a form)

"I'm sending data in JSON format" (the modern language of the web)

The actual message: "My name is Thomas"

The address of the service you're talking to

Extracting Archives

Extract .tar and .zip archives

`tar -czvf archive.tar.gz folder/` # Create

`tar -xvf archive.tar.gz` # Extract

`unzip archive.zip` # Extract .zip file

Option

-c

-x

-v

-f

-z

-j

-t

Meaning

Create a new archive

Extract files from archive

Verbose: show file names during processing

File: specify archive file name

Use **gzip compression** (for .tar.gz)

Use **bzip2 compression** (for .tar.bz2)

List files inside an archive (without extracting)

Extracting Archives

Demo:

```
mkdir testdir && echo "Hello" > testdir/hello.txt  
tar -czvf testdir.tar.gz testdir/  
tar -tvf testdir.tar.gz  
tar -xvf testdir.tar.gz
```

Alias

Alias command useful to create custom shortcuts for frequently used commands.

- Temporary:

```
alias ll='ls -lrt'
```

- Permanent:

Open ~/.bashrc and add this line: `alias ll='ls -lrt'`

Then close and reopen terminal OR `source ~/.bashrc`

Others examples:

```
alias projects='cd ~/project'
```

PATH variable

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Your own script (e.g. myscript.sh) only runs if you're in the same directory your \$PATH.

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But system commands like ls, cd, work from anywhere.

Because the shell automatically searches in all directories listed in \$PATH when you run a command without a path.

PATH variable

echo \$PATH

-> usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games

When you run a command:

- ➔ If it finds the command in one of those directories → it runs it
- ➔ If not → it shows “command not found”

PATH variable

demo:

Create a simple shell script
gedit hello.sh

```
#!/bin/bash  
echo "Hello!!."
```

Make the script executable
chmod +x hello.sh

Add the line to your .bashrc:

```
export PATH=$PATH:$HOME/XXXXXX  
source ~/.bashrc
```

Run it from ANYWHERE